Pressure-Volume Loop Shape as a Fatigue Health Indicator for Pressure-Only Proprioception in Shared-Manifold Soft Grippers: A Simulation Study

Author: Mergen Ulziibayar · NYU Tandon, Dept. of Mechanical & Aerospace Engineering

Status: Draft v1.1 (2026-07-03). **This is a simulation-only modeling study.** No physical experiments are reported; every result is synthetic and is labeled as such. Modeling assumptions are stated as choices, not as calibrated predictions for a physical actuator.

Venue & status (GO decision taken 2026-07-02). The submission hold is lifted: **GO for arXiv-first posting** (`cs.RO`, cross-list `eess.SY`), decided 2026-07-02 under the owner's delegated portfolio review. Rationale: the summer deliverable was defined as a *public* arXiv preprint; the manuscript is complete, verified against the study JSONs, and written for exactly this scope (sim-only honesty and the single-generator caveat are foregrounded); an unshipped finished preprint loses value every week while carrying no upside. Peer-review submission is a separate, later decision: **IEEE RoboSoft** at its next deadline cycle for the sim/method contribution; **RA-L** only once a physical bench campaign exists. The remaining action is the mechanical upload by the owner (arXiv account + possible first-submission endorsement) — see `docs/SUBMISSION.md` for the checklist.

Abstract

Soft pneumatic grippers face two coupled deployment problems: silicone chambers fail by cyclic fatigue, and recovering gripper pose without added sensors relies on the pressure signals already in the control loop. We study, in simulation, whether the in-loop pressure–volume (P-V) hysteresis loop can serve as a *health indicator* of fatigue-driven proprioception degradation in the practical shared-manifold topology, where chambers share one supply to cut valve count. We build a transparent, analytically invertible pipeline: a viscoelastic (standard-linear-solid) pneumatic plant producing rate-dependent P-V loops; a canonical fatigue model (Mullins softening with partial recovery, irreversible compliance drift, late leak growth); fatigue-coupled piecewise-constant-curvature (PCC) tip kinematics; a seeded sensor model; and a shared-manifold-versus-isolated-supply network simulator in which inter-chamber cross-talk lives in the actuation-band dynamics. From 2,000 labeled traces (20 actuators × 5 life stages × 2 supply topologies × 2 contact states × 5 repetitions), split by **actuator identity**, we evaluate static and dynamic pressure-only correctors and four recalibration policies (never, cycle-count clock, P-V-triggered, always-on).

We report three findings, including a negative one. **(1)** A predicted advantage of a dynamic (lagged-input) corrector over a static ridge map under the shared manifold **does not materialize**: the cross-talk is real in the network dynamics but second-order, and the dynamic corrector improves pose error by ≈0% at physically reasonable parameters. **(2)** The dominant pressure-only proprioception error is instead the fatigue **compliance-scale drift**, which is topology-independent and causes a young-calibrated estimator's curvature error to grow by roughly two orders of magnitude over life. **(3)** The observable P-V loop area is a strong health indicator of that drift (Pearson r = 0.885, 95% CI [0.835, 0.958] on held-out actuators; per-actuator r median 0.957, range [0.9570, 0.9574], reported for structure only), and a P-V-health-triggered recalibration policy holds pose error within a stated accuracy budget at **60% fewer recalibrations** than an always-on policy (2 vs 5 per actuator), with the trigger threshold selected on training actuators only. A lead-time audit found **no positive temporal lead** under that deployed threshold on the 5-stage grid: the trigger crossing occurred at life 0.83, while the fixed-calibration budget crossing occurred earlier (median

0.71, range 0.56-0.71), giving median lead = -0.123 normalized life (6/6 held-out actuators nonpositive). A sensing-free cycle-count schedule tuned by the same rule fails that budget on held-out actuators (0.21 mm vs the 0.159 mm budget) at a similar recalibration count — the measured P-V state, not the mere passage of cycles, is what transfers across actuators. **Scope caveat (stated up front):** all 20 actuators are drawn from a single generative model, so "generalization to unseen actuators" tests robustness to that generator, not real device-to-device variation; and absolute pose errors remain sub-millimeter throughout, so the operational value of triggered recalibration grows with tighter accuracy requirements.

---

1. Introduction

Soft pneumatic grippers are attractive for safe, compliant manipulation, but two monitoring problems limit their deployment. **Fatigue:** silicone actuators fail by cyclic fatigue, and current practice detects failure only after rupture or via before/after diagnostics [Mosadegh 2014; Wong 2026]. **Proprioception:** recovering pose without added sensors uses the pressure signals already present in the control loop [L. Wang 2020]. Both problems are usually studied in isolation and, more importantly, under an *independent-chamber* assumption.

The most practical multi-finger topology shares a pneumatic manifold across chambers to reduce valve count and onboard mass [McDonald 2021; Moran 2024]. On a shared manifold, driving one chamber's valve perturbs the shared supply node and therefore its neighbors — inter-chamber **cross-talk**. This paper asks whether, in that topology, fatigue and proprioception are *physically coupled*: as a chamber fatigues its compliance rises, which (a) changes the P-V loop shape and (b) could change the cross-talk, so that a single pressure stream might monitor both health and pose.

We test this coupling in a deliberately transparent simulation, before any hardware. The modeling choices are made for analytic checkability (noise-free pose is recoverable to machine precision; isolated supplies give numerically zero cross-talk), and every estimator is validated against its own ground truth. The contribution of this paper is therefore the **pipeline, the design outputs, and a pre-specified modeling result framed as a prediction** (frozen in `docs/result_spine.md`, commit `d856455`, before result write-up) — not an experimental claim.

2. Related work and prior-art boundary

P-V hysteresis and fatigue. P-V hysteresis is *already* used as a before/after fatigue diagnostic: Mosadegh et al. [2014] assess PneuNet durability across >10^6 cycles by comparing P-V curves, and US Patent 10,639,801B2 [Mosadegh, Shepherd, Whitesides 2020] documents the same before/after protocol. Libby et al. [2023] show the loop shifts with fatigue, and report FEM-agreement degradation (96%→80%). Hysteresis-loop area as a damage proxy has strong cross-domain precedent — metals [Haghshenas 2021], flight-control health indicators [Guo 2021], SHM/acoustic-emission RUL [Galanopoulos 2023] — grounded in early-warning theory [Scheffer 2009] and the prognostics-pipeline / health-indicator-quality criteria of Lei et al. [2018]. **We therefore do not claim first use of P-V for fatigue.** The boundary is the cycle-resolved, operational health indicator and triggerable recalibration policy, with quantified trigger timing, separating irreversible drift from reversible Mullins recovery [Lavazza 2023; Liao 2021; Mars & Fatemi 2002].

Pressure-only proprioception. Multi-chamber pressure-only sensing exists [L. Wang 2020; L. Wang 2023; J. Wang 2025; Joshi & Paik 2023; Zou 2024; Preechayasomboon & Rombokas 2021], but assumes independent or clean per-chamber supply. The nearest collision is Lindenroth et al. [2023], which senses contact force on *coupled parallel* chambers but never frames inter-chamber cross-talk as a proprioception confound or links it to fatigue.

Shared-manifold modeling. Lumped resistance–compliance models of pneumatic circuits [Stanley 2021] provide the tool to model cross-talk as a function of chamber compliance; shared single-source architectures are common in practice [Moran 2024; McDonald 2021].

Correctors and recalibration. ARX/echo-state correctors and physical reservoir computing are established for pneumatic hysteresis [Jaeger & Haas 2004; Youssef 2022; Shen 2025]. Online adaptation to soft-sensor drift exists but is *always-on* [Kushawaha 2025; Thuruthel 2022], or masks anomalous sensors [Sugiyama 2025]; a *fatigue-health-triggered* recalibration is the contrast we evaluate.

Ground truth and evaluation. We follow the soft-proprioception evaluation conventions of Zhang et al. [2023] and report pose RMSE (contact state is generated and stored in the dataset but is not an estimation target in this study — the correctors regress curvature/pose only, so no contact F1 is reported); PCC [Webster & Jones 2010] is treated as a modeled output, never as measured truth.

3. Modeling and methods

The pipeline is implemented in Python (`sim/`, `pipeline/`, `scripts/`) with unit-tested gates at each stage. All randomness is seeded; the dataset is reproducible from a fixed seed and its integrity is pinned by a manifest SHA-256.

3.1 Viscoelastic plant and P-V loops (Phase A)

Chamber walls are modeled as a standard linear solid (Zener) in the pressure–volume analog: a spring k_1 in parallel with a Maxwell arm (k_2, relaxation time τ). This yields rate-dependent P-V hysteresis whose dissipation per cycle has the closed form $E_{loss}(\omega) = \pi\, k_2\, A^2\, (\omega\tau)/(1+(\omega\tau)^2)$, used as a ground-truth check on the numerical loop area, and reduces to the linear cross-talk model in the quasi-static limit. *(Validated: Phase A PASS.)*

3.2 Fatigue model (Phase B)

A canonical, deterministic life law injects: Mullins softening with a 30% permanent floor and 24-h recovery time constant; irreversible compliance drift; curvature-acceleration onset at 70% life; and late leak growth (to 20× conductance) observed by a separate closed-valve pressure-decay probe. Degradation enters the plant as a rising compliance multiplier (softer

k_1) and loss multiplier (larger k_2). (Validated: Phase B PASS, synthetic consistency.)

3.3 P-V health features (Phase C)

Causal P-V features, Mullins-recovery normalization, health-index metrics, and matched false-alarm detectors were validated across a 144-condition identifiability grid; a matched segmented model recovers the quadratic onset and fails on logistic-onset variants — the intended inverse-crime/generalization control. *(Validated: Phase C PASS, synthetic.)* Phase C metrics are applied, not re-validated, downstream.

3.4 Fatigue-coupled PCC kinematics (Phase D)

Realized chamber pressure maps to bending curvature through the fatigue compliance multiplier, $\kappa = \kappa\_gain \cdot C(\text{life}) \cdot \max(P - P_0, 0)$, so fatigue *amplifies* curvature at fixed pressure. Curvature maps to tip SE(3) by the robot-independent PCC convention [Webster & Jones 2010]. The map is analytically invertible: forward↔inverse round-trips to machine precision, and a noise-free pose is recoverable exactly (gate test).

3.5 Sensor model

A seeded measurement model applies Gaussian noise, quantization, decimation (sampling), and a contact observation (binary flag + synthetic normal force from tip penetration). Independent sub-streams per channel make the corruption reproducible and channel-order-independent.

3.6 Shared-manifold vs isolated network dynamics

A time-domain integrator drives a focal chamber plus two neighbors. In the **shared** topology all chambers draw from one supply node; driving one valve pulls the shared node and perturbs the neighbors. In the **isolated** topology each chamber has its own regulated node and the state blocks are decoupled. Consistent with a lumped-RC pre-test (Gate 0), the cross-talk is *dynamic*: its DC gain is compliance-independent while its actuation-band gain drifts with fatigue. Excitations are multisine in the 1–5 Hz actuation band so the effect is observable. Gate tests confirm: isolated coupling is numerically zero, shared coupling is measurable, and the DC-independent / actuation-band-drift property holds.

3.7 Dataset

20 actuators × 5 life fractions {0.1,0.3,0.5,0.7,0.9} × 2 topologies × 2 contact states × 5 repetitions = **2,000 traces** (240 time samples each). Per-actuator geometry, wall stiffness, and rupture life are drawn deterministically. The split is by **actuator identity** (14 train / 6 held-out), never by trace, so generalization is measured on actuators never seen in training. Features stored per trace include the shared-manifold pressure, all chamber valve commands, the focal chamber pressure and volume, the true and measured tip positions, the true curvature, and contact.

3.8 Proprioception correctors (Phase E)

The target is focal bending curvature (and, through PCC, tip position) from pressure-only features [shared-manifold pressure, all chamber commands]. We compare a **static ridge** map (memoryless) against a **dynamic** corrector that is the identical ridge estimator on lagged inputs (an exogenous ARX/FIR form; output feedback is excluded because the pose is the estimation target, not an online measurement). The lag count is the only difference.

3.9 P-V health signal and recalibration policies (Phase F)

The health signal is the **observable** P-V loop area from a volumetric probe (the loop-shape quantity, not the ground-truth compliance), normalized to its young value so the trigger is a fractional loop-area growth comparable across actuators. Three policies, each starting from one calibration at the youngest life stage: **fixed** (never recalibrate), **always** (recalibrate every stage), and **triggered** (recalibrate when fractional loop-area growth since the last calibration exceeds τ). The threshold τ is selected on **training actuators only** against a stated accuracy budget, then applied unchanged to held-out actuators. We report estimation error **and** recalibration count together, so a "savings" cannot hide degraded accuracy.

4. Results

4.1 Validation gates

All structural gates pass: noise-free pose recovery and forward↔inverse curvature round-trips to machine precision; isolated-supply off-diagonal coupling at the solver-noise floor ($<10^{-6}$) while shared coupling is measurable ($>10^{-4}$); and the Gate-0 property that DC cross-talk is compliance-independent while actuation-band cross-talk drifts with fatigue. The full unit-test suite gates these checks.

4.2 Cross-talk is real but second-order (negative result, retained; Fig. 1)

The proposal predicted that shared-manifold cross-talk drift would be visible to a dynamic corrector and invisible to a static ridge, making the dynamic corrector win under the shared manifold. **This does not occur.** With all chamber commands available as features and per-actuator calibration, the dynamic (lagged-input) corrector improves curvature RMSE over the static ridge by **≈0% under**

either topology (shared: 0.244 vs 0.244 1/m; isolated: 0.295 vs 0.295 1/m; pose ≈0.84 vs 0.84 mm and 1.02 vs 1.02 mm). The cross-talk is genuine in the network dynamics (§4.1) but is a small perturbation (≈4% of the focal curvature at these physically reasonable parameters) that does not materially degrade pose estimation. We retain this as a negative result and did not tune network parameters to manufacture the predicted effect; §5 characterizes the envelope over which it holds by sweeping the supply softness.

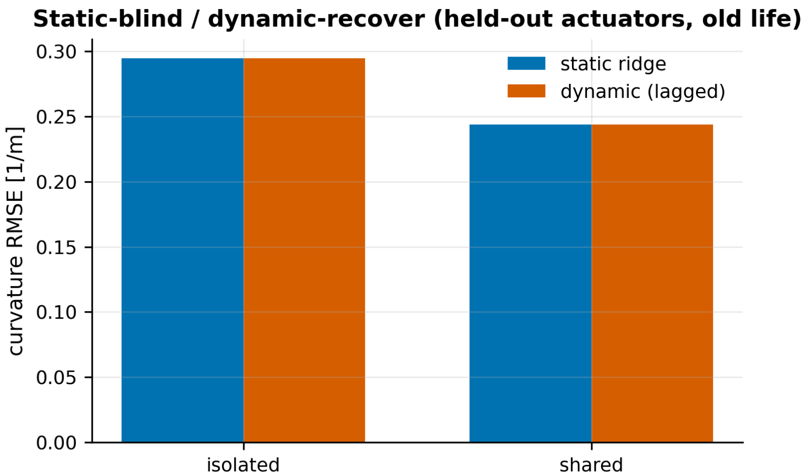

Figure 1. Static vs dynamic (lagged-input) corrector on held-out actuators at old life; the dynamic corrector yields ~0% improvement under either supply topology — the predicted shared-manifold cross-talk advantage does not appear.

4.3 Compliance drift dominates the proprioception error (Fig. 2)

A young-calibrated static estimator's curvature RMSE grows by roughly two orders of magnitude over life (≈0.004 → ≈0.46–0.56 1/m at 0.9 life), and the shared and isolated curves are comparable — the difference does not consistently favor the shared-manifold-degrades-more hypothesis. The dominant pressure-only proprioception error is therefore the **topology-independent fatigue compliance-scale drift**, not cross-talk. This both explains §4.2 and motivates recalibration as the right intervention.

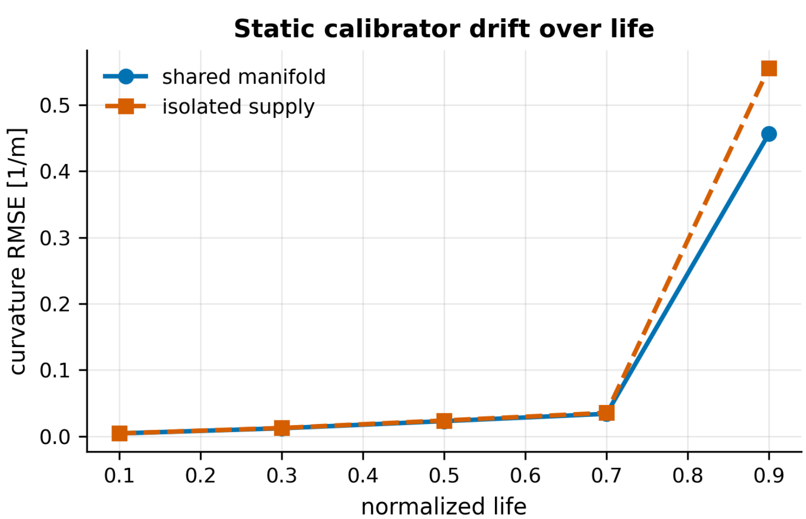

Figure 2. Young-calibrated static estimator curvature RMSE over normalized life (shared vs isolated supply); error grows ~100x by end of life and is comparable across topologies — the degradation is the topology-independent compliance drift, not cross-talk.

4.4 P-V loop area is a health indicator, not a positive-lead trigger on this grid (Fig. 3)

On held-out actuators, the observable P-V loop-area fractional growth tracks the fixed-calibration pose error over life with **Pearson *r* = 0.885, 95% CI [0.835, 0.958]** (bootstrap, 2,000 resamples; CI excludes 0). The same monotone structure also appears within actuators: per-actuator *r* has median 0.957 and range [0.9570, 0.9574] across the six held-out actuators (5 stages each; reported for structure consistency, not significance).

The temporal lead audit is negative. Using the deployed train-selected τ*=0.05 threshold, the trigger crossing occurs at life 0.83 for each held-out actuator, while the young/fixed-calibration error crosses the 0.159 mm budget earlier: median life 0.71, range 0.56-0.71. Linear interpolation between the five life stages gives median lead = -0.123 normalized life, range [-0.269, -0.115] (about -972 to -375 cycles across the held-out rupture lives); all 6/6 held-out actuators are nonpositive-lead cases. Thus the observable loop area is a strong health correlate and useful recalibration trigger, but this 5-stage study does **not** demonstrate positive temporal lead.

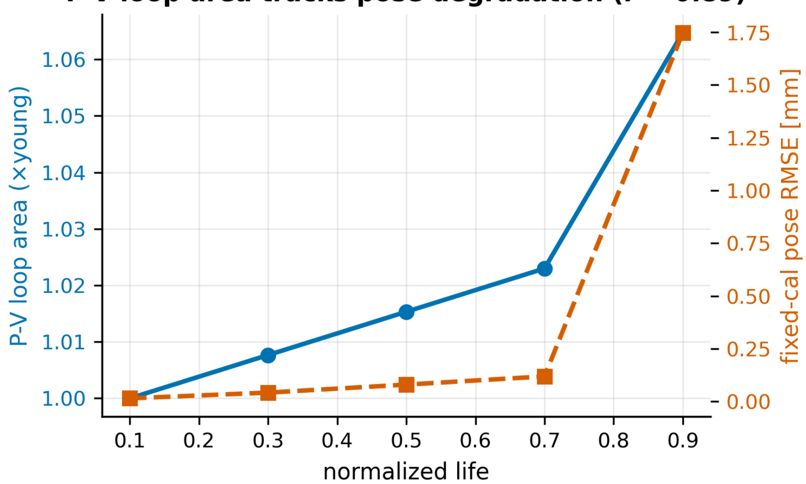

Figure 3. Observable P-V loop-area growth (left axis) tracks fixed-calibration pose error (right axis) over life on held-out actuators; bootstrap r = 0.885, 95% CI [0.835, 0.958]. The deployed trigger threshold does not provide positive temporal lead on the 5-stage grid.

4.5 Recalibration trade-off (Fig. 4)

Against a stated 0.159 mm accuracy budget (selected on training actuators as halfway from the always-on error toward the never-recalibrate error), we compare four policies. Besides fixed and always-on, the P-V trigger must beat the obvious *sensing-free* competitor: a **cycle-count clock** that recalibrates every *T* actuation cycles. Both adaptive policies are tuned on training actuators by the same rule (fewest recalibrations within budget): τ\ = *0.05 fractional loop-area growth for the trigger,* T\ = 2,700 cycles for the clock (train errors 0.044 and 0.132 mm respectively — both within budget on train). On held-out actuators:

Policy	Pose RMSE (held-out)	Recalibrations / actuator	Meets 0.159 mm budget?
fixed (never)	0.40 mm	1	no
cycle-count clock (*T**=2,700)	0.21 mm	1.5	**no — fails on transfer**

Policy	Pose RMSE (held-out)	Recalibrations / actuator	Meets 0.159 mm budget?
P-V-triggered (τ*=0.05)	**0.06 mm**	**2**	**yes (2.7× margin)**
always-on	0.02 mm	5	yes

Two comparisons matter. Against always-on, the trigger holds pose error near the always-on band at **60% fewer recalibrations** (2 vs 5 per actuator). Against the clock — the comparison that isolates what the *P-V measurement itself* buys — the clock, tuned within budget on training actuators, **violates the budget on held-out actuators** (0.21 mm > 0.159 mm) at a similar recalibration count, because a global cycle period misaligns with per-actuator degradation state whenever rupture life varies across devices (here drawn from 3,000–4,000 cycles), while the trigger reads that state directly and transfers safely. Absolute errors are sub-mm in all policies (see §5), so the demonstrated value is the error-vs-cost trade-off and its transfer to unseen actuators, not the absolute accuracy.

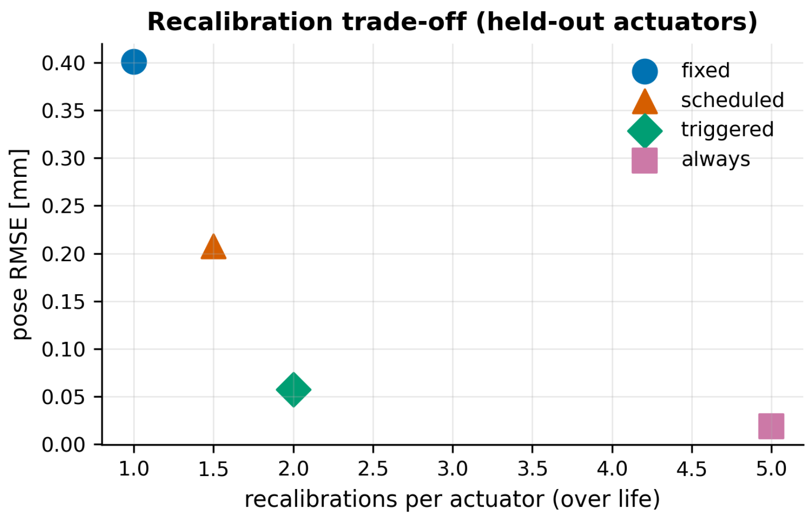

Figure 4. Recalibration trade-off on held-out actuators — pose error vs recalibrations per actuator for fixed, cycle-count clock, P-V-triggered, and always-on policies. The triggered policy meets the accuracy budget at 60% fewer recalibrations than always-on; the sensing-free clock, tuned within budget on training actuators by the same rule, fails the budget on transfer.

5. Discussion and limitations

• **Simulation only.** Every result is synthetic. The pipeline and the modeling result (framed as a prediction) are the contributions; no experimental validation is claimed.

• **One generative model.** All 20 actuators come from a single generator, so held-out evaluation tests robustness to that generator's variation, not real device-to-device spread. This is the first thing a reviewer should weigh and is stated in the abstract.

• **What could have made *r* low (why 0.885 is informative and not built in).** A fair objection: the fatigue model makes both compliance drift and loop area monotone in the same latent life variable, so doesn't the correlation follow by construction? Between the latent state and the reported statistic sit the mechanisms the study exists to test, and each could have broken it: the correlation uses the **observable** loop area from a noisy, quantized, decimated volumetric probe at one drive frequency — not the ground-truth compliance — so probe noise and the finite excitation could have decorrelated it; **Mullins recovery** superimposes a reversible, rest-history-dependent loop-area component on the irreversible drift, and a probe that failed to separate them would track rest schedule, not damage; the pose error depends on the *estimator through the PCC map and sensor chain*, not on compliance directly, so a poorly conditioned corrector could have decoupled error growth from loop-area growth; and per-actuator variation in geometry, wall stiffness, and rupture life (drawn independently) scatters the pooled correlation — the same variation that defeats the cycle-count clock in §4.5. The CI half-width (±0.06) measures how much these mechanisms *did* degrade the link. What the single-generator caveat still owns: real silicone adds failure modes (delamination, local tearing, temperature sensitivity) that no parameter draw from this generator produces — which is exactly what the hardware spot-check must test.

- **The headline cross-talk hypothesis was not supported.** Cross-talk is monotone in compliance

(Gate 0) but second-order for pose at realistic parameters. The proposal anticipated this "benign coupling" outcome as a publishable floor; the deliverable is the health-indicator + recalibration result via the compliance-drift pathway.

- **Parameter-sensitivity of cross-talk (envelope, Fig. 5).** To make the "second-order" claim a

characterized regime rather than a single-point assertion, we swept the network parameters that set the coupling magnitude and measured the neighbor/driven response ratio at the actuation band (`probe_coupling`, `scripts/run_study4.py`). The coupling rises monotonically with supply softness: at the default parameters it is **6.3%**, and it crosses the **10%** "starts to matter" line only when the supply resistance R_s is **≈1.7×** softer and the **20%** line at **≈4.3×** softer. The manifold compliance C_m is a far weaker knob — a larger buffer reduces coupling, and across a 128× span (×0.25 to ×32) it stays in the **5.8–6.3%** band and never reaches 10%. So the negative result is not knife-edge: cross-talk remains second-order across realistic shared-manifold designs and would only become first-order for pose under a deliberately under-provisioned (several-fold throttled) supply. A confirmatory corrector check at the softer settings is consistent — the dynamic (lagged) corrector still does not beat the static ridge at these small-sample sizes, though its penalty shrinks as coupling grows.

- **Sub-millimeter absolute errors.** Even never-recalibrating yields ≈0.40 mm pose RMSE at this

sensor-noise level, so the operational case for triggered recalibration strengthens with tighter pose requirements, softer actuators (larger per-stage drift), or higher noise.

- **Modeling choices.** The fatigue law, PCC single-segment kinematics, and lumped network are

transparent assumptions chosen for analytic checkability, not calibrated fits.

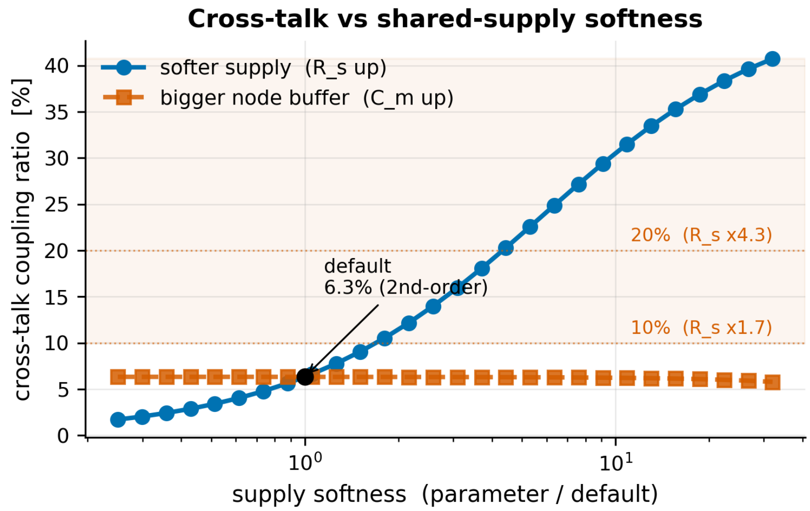

Figure 5. Cross-talk coupling ratio (neighbor/driven response at the actuation band) vs shared-supply softness, as a multiple of each default parameter. Coupling rises monotonically with supply resistance R_s and is ≈6.3% at the default operating point (marked); it reaches the 10% and 20% "first-order" lines only at R_s ≈1.7× and ≈4.3× softer. Manifold compliance C_m is a weak knob (stays 5.8–6.3% across ×0.25–×32). The negative cross-talk result holds across realistic shared-manifold designs.

6. Conclusion

In simulation, the observable P-V loop shape is a quantified health indicator (r ≈ 0.89) of fatigue-driven pressure-only proprioception drift in shared-manifold soft grippers, but the deployed threshold does not show positive temporal lead on the 5-stage grid. A P-V-health-triggered recalibration policy meets an accuracy budget at a fraction of the recalibration cost of always-on adaptation, transferring from training to held-out actuators — where a sensing-free cycle-count schedule tuned by the same rule does not. The shared-manifold cross-talk, while real, is second-order for pose at realistic parameters — a retained negative result. The natural next step is a physical bench campaign to anchor the synthetic story (single-actuator micro-tear spot-check, then the N≈10 fatigue campaign of the companion proposal).

7. Reproducibility

All code, the dataset generator, and the analysis scripts are in the repository. The dataset is regenerable from a fixed seed (`python -m scripts.phaseD_dataset`; integrity pinned by the manifest SHA-256); the studies are `python -m scripts.run_study2` (correctors), `python -m scripts.run_study3` (health indicator + recalibration), and `python -m scripts.run_study4` (cross-talk parameter-sensitivity envelope). Frozen result JSON and figures are committed under `data/sim/phaseD/`; the pre-specified claim and target figures are in `docs/result_spine.md` (commit `d856455`, before result write-up). The full unit-test suite and manuscript-number check gate the pipeline in CI.

References

Citations follow the annotated literature base (`docs/A01_A04_Literature_Review.md`), where full DOIs are listed. Selected works cited here:

- Mosadegh et al. 2014, *Adv. Funct. Mater.* 24(15):2163–2170. DOI 10.1002/adfm.201303288.
- Mosadegh, Shepherd, Whitesides 2020, US Patent US10639801B2.
- Libby et al. 2023, *ISMR 2023*. DOI 10.1109/ISMR57123.2023.10130227.

- Wong, Luo, Scharff 2026, *Adv. Robotics Research* e202500172. DOI 10.1002/adrr.202500172.
- Mars & Fatemi 2002, *Int. J. Fatigue* 24(9):949–961. DOI 10.1016/S0142-1123(02)00008-7.
- Lavazza, Contino, Marano 2023, *Mech. of Materials* 178:104560. DOI 10.1016/j.mechmat.2023.104560.
- Liao et al. 2021, *Int. J. Mech. Sci.* 206:106624. DOI 10.1016/j.ijmecsci.2021.106624.
- Scheffer et al. 2009, *Nature* 461:53–59. DOI 10.1038/nature08227.
- Lei et al. 2018, *MSSP* 104:799–834. DOI 10.1016/j.ymssp.2017.11.016.
- Haghshenas, Jang, Khonsari 2021, *Mech. of Materials* 155:103734. DOI 10.1016/j.mechmat.2020.103734.
- Guo et al. 2021, *Measurement* 186:110076. DOI 10.1016/j.measurement.2021.110076.
- Galanopoulos et al. 2023, *Eng. Structures* 290:116391. DOI 10.1016/j.engstruct.2023.116391.
- Sugiyama et al. 2025, *Front. Robot. AI* 11:1504651. DOI 10.3389/frobt.2024.1504651.
- L. Wang & Z. Wang 2020, *Soft Robotics* 7(2):198–217. DOI 10.1089/soro.2018.0135.
- L. Wang et al. 2023, *Soft Robotics* 10(4):825–837. DOI 10.1089/soro.2021.0056.
- J. Wang et al. 2025, *Adv. Intell. Syst.* 7(4):2400534. DOI 10.1002/aisy.202400534.
- Joshi & Paik 2023, *Soft Matter* 19. DOI 10.1039/D2SM01197B.
- Zou et al. 2024, *Nature Communications* 15:539. DOI 10.1038/s41467-023-44517-z.
- Preechayasomboon & Rombokas 2021, *Actuators* 10(2):30. DOI 10.3390/act10020030.
- Lindenroth et al. 2023, *IEEE/ASME T. Mechatronics* 28(1):80–91. DOI 10.1109/TMECH.2022.3210065.
- Moran et al. 2024, *Comms. Eng.* 3:117. DOI 10.1038/s44172-024-00251-y.
- McDonald & Ranzani 2021, *Front. Robot. AI* 8:720702. DOI 10.3389/frobt.2021.720702.
- Stanley et al. 2021, *ASME J. Dyn. Sys. Meas. Control* 143(5):051001. DOI 10.1115/1.4049009.
- Jaeger & Haas 2004, *Science* 304(5667):78–80. DOI 10.1126/science.1091277.
- Youssef et al. 2022, *Micromachines* 13(2):216. DOI 10.3390/mi13020216.
- Shen, Miyazaki, Kawashima 2025, *RA-L*, arXiv:2409.06961.
- Kushawaha et al. 2025, arXiv:2503.16540.
- Thuruthel, Gardner, Iida 2022, *Soft Robotics* 9(6):1167–1176. DOI 10.1089/soro.2021.0012.
- Webster & Jones 2010, *IJRR* 29(13):1661–1683. DOI 10.1177/0278364910368147.
- Zhang et al. 2023, *IROS 2023*:2564–2571. DOI 10.1109/IROS55552.2023.10342379.